

SHRUTI MANDAVKAR

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PROFESSIONAL SUMMARY

High-impact Data & Observability Engineer with comprehensive experience architecting high-throughput real-time ingestion pipelines, time-series telemetry stacks, spatial analytics, and AI/ML system integrations. Proven expertise in engineering tier-1 enterprise-grade monitoring solutions for global industrial clients (Hitachi, BKT Tires, Boldrocchi, CST, Mitsuba) and complex geological datasets using Prometheus, VictoriaMetrics, Grafana, Plutono, and PostgreSQL/TimescaleDB. Adept at containerizing microservices, optimizing GenAI/RAG retrieval pipelines, and delivering scalable data infrastructure for high-concurrency environments.

TECHNICAL SKILLS

- Observability & Visualization:** Grafana, Plutono, VictoriaMetrics, Prometheus, Geomaps, Time Series, Datagrid Views, Custom Alerting Engine
- Databases & Data Storage:** PostgreSQL, TimescaleDB, MongoDB, MySQL, Redis, pgAdmin, DBeaver, Relational Schema Design & Indexing
- Data Engineering & IoT:** Real-Time Telemetry Ingestion, MTConnect Protocol (CNC Integration), REST APIs, FastAPI, Node.js, Express.js, QGIS (Spatial Vectors)
- Programming & Web:** Python, SQL, C Programming, Java (Core), JavaScript, R, HTML5/CSS3, Flutter (Cross-Platform Remote Dashboards)
- AI & Machine Learning:** RAG Pipeline Optimization, Generative AI (LLMs / GPT-4), Feature Engineering, Anomaly Detection, Natural Language Processing (NLP)
- DevOps & Infrastructure:** Docker (Multi-Stage Containerization), Git, GitHub, Linux System Administration, CI/CD Workflows, Cybersecurity Essentials

PROFESSIONAL EXPERIENCE

Research Intern – Data & Observability Engineering | IIT Bombay (Machine Intelligence Group - NCAIR) Mumbai, India | June 2025 – Present

- Enterprise Observability Stacks:** Architected and deployed production-grade real-time telemetry dashboards on Plutono and Grafana for Tier-1 industrial enterprises including Hitachi, BKT Tires, CST, Boldrocchi, and Mitsuba, accelerating operational insights.
- Campus-Wide Data Stack:** Engineered a campus-wide observability architecture for IIT Bombay Camera Status Data using VictoriaMetrics and Grafana, integrating automated alerting, time-series visualizations, datagrids, and interactive Geomaps.
- Automated Ingestion Pipelines:** Engineered automated data cleaning and ingestion pipelines in PostgreSQL to process raw high-frequency industrial telemetry, designing interactive monitoring dashboards for pipeline integrity.
- IoT & CNC Machine Integration:** Containerized IoT data ingestion engines using Docker and MTConnect API protocols, writing low-level C functions to stream high-frequency CNC machine telemetry live into Plutono dashboards.
- Time-Series Query Optimization:** Optimized high-throughput time-series workflows using TimescaleDB, PostgreSQL, pgAdmin, and DBeaver to store, index, and query continuous sensor streams with minimal execution latency.
- AI / RAG Pipeline Optimization:** Diagnosed and eliminated architectural bottlenecks in internal Retrieval-Augmented Generation (RAG) pipelines, significantly improving response latency and contextual retrieval accuracy.
- Industrial ML Analytics:** Applied feature extraction, normalization, and machine learning preprocessing techniques to industrial mine telemetry for Project Dhatuverse.
- Mobile & DevOps Integration:** Developed a cross-platform Flutter application for real-time remote equipment metric monitoring, integrated with Docker containers and GitHub CI/CD pipelines.

Data Analyst Intern | Genesys International Corporation Ltd. Mumbai, India | Dec 2024 – Apr 2025

- Real-Time Spatial APIs:** Architected high-throughput RESTful spatial data ingestion APIs using FastAPI to handle concurrent streaming feeds for a live Crime Analytics Dashboard.

- **Geological & PromQL Analytics:** Wrote complex PromQL and SQL analytical queries on high-density geological datasets across Maharashtra and Karnataka, rendering real-time regional insights via Grafana dashboards.
- **Infrastructure Observability:** Established centralized infrastructure monitoring frameworks by integrating Prometheus time-series metrics and PostgreSQL system logs into dynamic Grafana health dashboards.
- **GIS Spatial Data Processing:** Leveraged QGIS for spatial data cleaning, processing, and vector transformations, seamlessly integrating geographic vectors into live monitoring workflows.

KEY TECHNICAL PROJECTS

FieldWisdom: Production-Grade Industrial AI & Observability Platform | Full-Stack / ML / IoT

Mumbai, India | 2025 - 2026

- Engineered a containerized full-stack platform using Node.js, Express, and MongoDB document storage to streamline real-time knowledge and telemetry retrieval for field engineers.
- Built data ingestion pipelines feeding an ML intelligence layer for field telemetry anomaly detection and semantic audio/text processing.
- Designed multi-stage Docker builds serving production asset bundles to eliminate container latency and reduce resource footprint.

AI Resume Matcher & Microservices Observability Stack | GenAI / Microservices

Mumbai, India | 2025

- Engineered an automated data extraction and parsing engine utilizing NLP and GPT-4 / LLM APIs to match candidate qualifications against job descriptions.
- Deployed a resilient microservices architecture using Docker and reverse proxies, configuring full-stack observability with Prometheus metrics and Grafana alerting to monitor API latency and system health.

EDUCATION

Bachelor of Engineering (B.E.) in Information Technology | A. P. Shah Institute of Technology (APSIT)

Mumbai, India | 2026

- Core Specialization: Data Science, Machine Learning, Data Engineering, Cloud Systems & Systems Observability.

Higher Secondary Certificate (H.S.C. - Science) | St. Xavier's High School & Jr. College

Mumbai, India | 2021

- Academic Achievement: Scored 78% in Science Stream.

CERTIFICATIONS & KEY ACHIEVEMENTS

- **Technical Certifications:** AWS Academy Data Analyst Certification | Core Java Certification | AI Job Simulation (Forge)
- **Awards & Honors:** Silver Medalist - Technical Poster Presentation | 1st Prize - NCC Training Exhibition | District-Level Science Exhibition Participant